

Songhan Hu

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EDUCATION

Harvard University

Major: Master's Degree of Data Science, MIT Cross-Registered Student

GPA: 3.97/4.0

Cambridge, MA

Aug 2024 – May 2026

University of California, Irvine (UCI)

Major: Mathematics (Concentration in Finance), Double: Quantitative Economics

GPA: 3.95/4.0 (Rank 1/160), Double Magna Cum Laude

Irvine, CA

Sep 2020 - June 2024

SKILLS

Software: Python, SQL, MATLAB, R, Stata, LATEX, Git, Linux, Tableau, and GCP

Frameworks: Scikit-learn, TensorFlow, Keras, PyTorch, Pandas, NumPy, SciPy, Matplotlib, Seaborn and FastAPI

Relevant Techniques: Statistical Regressions Analysis, Mathematical Modeling, Machine Learning, LLM and CV

RESEARCH AND ACADEMIC PROJECTS

Non-Standard Jade Identification & Explanation AI

Sep 2025 – Now

- Engineered an end-to-end **scalable data pipeline** for 2M+ jade product listings from huiliu.com, transforming unstructured metadata, livestream subtitles, and multimedia files into machine-learning-ready datasets with automated validation, deduplication, and feature extraction.
- Developed a **Mixture-of-Experts (MoE)** pricing model integrating Qwen multimodal embeddings (image + text) to predict market values, incorporating a **Price Inflation Adjustment Neural Network** module to mitigate data imbalance and achieving $RMSE \approx \text{¥}1,000$.
- Implemented an **image-text RAG** system leveraging pgvector-based semantic retrieval and **instruction-tuned GPT** backbones to generate interpretable pricing rationales, enhancing explainability and trust for non-standard goods.
- Deployed the complete system on **Vertex AI and Cloud Run**, linking a user-facing web interface to backend microservices that process user inputs through asynchronous feature extraction, multimodal embedding, and MoE inference, with semantic caching, achieving **<1 s latency and >99% uptime** under concurrent user loads.

Soybean Future Price Forecasting

Feb 2025 – May 2025

- Designed a multi-source forecasting pipeline (daily returns + design-ready CFTC positioning module with weekly forward-fill & release-day dummies) for next-day direction
- Implemented and customized **TimesNet**, adding a **composite loss** to emphasize directional accuracy and magnitude awareness; tuned hyperparameters with **Bayesian search** + early stopping
- Calibrated probabilities converted to position sizes via **Kelly scaling** with risk caps; fusion model **backtests** delivered **Sharpe 1.64** out-of-sample (2023–24)

Real-Time NBA Game Outcome Prediction System

Sep 2024 – Dec 2024

- Developed a real-time NBA prediction model integrating player SCAB scores, team synergy, and betting odds market signals (i.e. betting odds jump density)
- Achieved **84.1% accuracy**, surpassing Elo benchmark by 33%, validated against historical outcomes and live betting odds
- Built **CI/CD** pipelines with unit tests and GitHub integration for robust, **automated deployment**

PROFESSIONAL EXPERIENCE

Bank of America | Quantitative Strategist Intern – Linear Rates

New York, NY/ June 2025 - Aug 2025

- Pioneered **regression-swap** bootstrapping on non-standard SOFR nodes (7Y/15Y/20Y); compressed pricing errors by **6 bps** and delivered **+6%** improvement in hedge **DV01 accuracy** vs. vanilla bootstraps; drafted a 50Y regression anchor to retain structure while redistributing delta to 5Y/10Y/30Y
- Built a scoring system that averages two orthogonal metrics: (1) synthetic delta-hedge portfolio PnL and (2) implied-vs-real fly spread stability (time-series diagnostics). The score auto-selects the daily curve for production and delivered **–20% fly variance** with tighter hedge residuals
- Designed multiple regression variants—**ridge**, rolling look-back windows, and **time-decay weights**—and performed **grid search** to calibrate coefficients & update frequency (dynamically)
- Developed a Python engine to **automate** curve construction, walk-forward backtests, scoring, and reporting; standardized data cleaning/day-count conventions; generated **dashboards**

Shenwan Hongyuan Securities | Quantitative Researcher Intern

Shanghai, China/ June 2023 - Aug 2023

- Engineered a **dynamic sector rotation strategy** using four-point momentum factors; achieved Sharpe ratio of 1.5 through factor construction and systematic backtesting
- Built an **ETF selection model** leveraging industry-level PE ratios; applied **mean-variance optimization** to dynamically adjust weights under risk constraints
- Designed and deployed a Fund of Funds (FoF) strategy using a **random forest model** to predict fund returns; delivered 30% annualized alpha over benchmark with a Sharpe ratio of 1.2

China International Capital Corporation | Part-time Quantitative Analyst

Remote/ Jul 2022 - Aug 2022

- Performed statistical analysis using the **T-M model** to evaluate stock selection and timing ability for the GF multi-factor mixed fund, contributing to published annual performance reports
- Improved decision-making by creating 200+ data-driven **visualizations**, analyzing industry allocation, and tracking daily returns across 74 target funds, which informed portfolio allocation adjustments